## 6.7 - L'Hopital's Rule

**Goals:** Recognize different types of indeterminate forms and limits using L'Hospital's Rule and
**Before Class Videos:** 4.5.1, 4.5.2

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Which of the following are indeterminate forms?

| | | |
|---|---|---|
| $\frac{\infty}{\infty}$ | $\frac{\infty}{0}$ | $\frac{0}{\infty}$ |
| $\frac{0}{0}$ | $0 \cdot \infty$ | $\infty \cdot \infty$ |
| $\infty + \infty$ | $\infty - \infty$ | $0^0$ |
| $1^\infty$ | $0^\infty$ | $\infty^0$ |

**Case 1: Quotient Forms**

**Theorem (L'Hospital's Rule):**
If $f$ and $g$ are differentiable functions on an open interval $I$ containing $a$ and $g'(x) \neq 0$ on $I$ except possibly at $c$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

Note: $a$ can be $\pm\infty$ too, and the can be $\pm\infty$ too, and the limits can be one sided

**Example 1:**

$$\lim_{x \to 0} \frac{\sin x}{x}$$

**Example 2:**

$$\lim_{x \to 0} \frac{e^x - x - 1}{x^2}$$

**Example 3:**

$$\lim_{x \to 0^+} \frac{\ln(e^x - 1)}{\ln x}$$

**Case 2: Exponential Forms.**

1. Instead of $f(x)^{g(x)}$, study $\ln(f(x)^{g(x)})$.
2. Apply the log rule to get $g(x)\ln(f(x))$.
3. Rewrite as a quotient.
4. Apply Case 1 L'Hospital's Rule.
5. Take the exponential of the final value.

**Example 4:**

$$\lim_{x \to \infty} x^{e^{-x}}$$

**Example 5:**

$$\lim_{x \to e^+} (\ln x)^{1/(x-e)}$$

**Example 6:**

$$\lim_{x \to (\pi/2)^-} (\tan x)^{\cos x}$$

**Case 3: Products and Differences like $\infty \cdot 0$, $\infty - \infty$, etc.**

Use algebra to convert to a $0/0$ or $\infty/\infty$ form, then apply Case 1.

**Example 7:**

$$\lim_{x \to (\pi/2)^-} \left(\frac{\pi}{2} - x\right) \tan x$$

**Example 8:**

$$\lim_{x \to 0^+} \left(\frac{3x+1}{x} - \frac{1}{\sin x}\right)$$

**Example 9:**

$$\lim_{x \to 0^+} \left(\frac{3x-1}{x} - \frac{1}{\sin x}\right)$$

**Example 10:** Final combo example:

$$\lim_{x \to 0^+} (\sin x)^{1/\ln x}.$$

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**Suggested after class problems** from APEX: Section 6.7: 11,24,35,39